

- RAG Learning

Molecules, Transport and Health

Edexcel A A-Level Biology — Topic 1

1.1 Water

Water: A polar inorganic molecule (H_2O) that acts as the universal solvent in biological systems. It is essential for metabolic reactions, transport, temperature regulation and as a habitat.

Molecular Structure of Water

Water is a simple molecule consisting of one oxygen atom covalently bonded to two hydrogen atoms. The bond angle is approximately 104.5 degrees. Because oxygen is significantly more **electronegative** than hydrogen, the shared electrons in each O–H bond are pulled towards the oxygen atom. This creates a **permanent dipole**: the oxygen carries a partial negative charge (δ^-) and each hydrogen carries a partial positive charge (δ^+).

This polarity enables water molecules to form **hydrogen bonds** with each other. Each water molecule can form up to four hydrogen bonds with neighbouring molecules (two through its hydrogen atoms acting as donors, two through its oxygen lone pairs acting as acceptors). Although each individual hydrogen bond is weak (approximately 1/20th the strength of a covalent bond), collectively they give water its remarkable properties.

Properties of Water and Their Biological Significance

Property	Molecular Explanation	Biological Importance
Excellent solvent	The partial charges on water molecules attract ions and polar molecules, surrounding	Most metabolic reactions occur in aqueous solution; transport medium in blood plasma, lymph, xylem sap and

	them and separating them from each other (hydration shells)	cytoplasm; ions (Na^+ , K^+ , Cl^-) dissolve readily for nerve impulses and muscle contraction
High specific heat capacity ($4.18 \text{ J g}^{-1} \text{ }^\circ\text{C}^{-1}$)	A large amount of energy is needed to break the many hydrogen bonds before temperature rises	Resists temperature fluctuations, providing a stable environment for enzyme-catalysed reactions; aquatic habitats remain relatively constant; important for thermoregulation in endotherms
High latent heat of vaporisation (2260 J g^{-1})	Significant energy is required to break hydrogen bonds to convert liquid water to vapour	Effective cooling mechanism — evaporation of sweat removes heat from the body surface; transpiration cools plant leaves
Cohesion and adhesion	Cohesion: hydrogen bonds hold water molecules together. Adhesion: water attracted to other polar surfaces	Cohesion enables water columns in xylem to be pulled upward without breaking (cohesion-tension theory); surface tension allows small insects to walk on water; adhesion helps water climb narrow tubes (capillary action)
Maximum density at 4°C (ice is less dense than liquid water)	In ice, hydrogen bonds hold molecules in a rigid, open crystalline lattice with more space between molecules	Ice floats on the surface of ponds and lakes, insulating the water below and allowing aquatic organisms to survive winter; if ice sank, entire water bodies would freeze solid from the bottom up
Transparency	Visible light passes through water	Allows photosynthesis to occur in aquatic environments at depth; light reaches aquatic plants and algae
Metabolic role	Water is a reactant and product in many reactions	Hydrolysis reactions (digestion) require water; condensation reactions release water; photolysis of water in photosynthesis provides electrons and protons

Exam Tip: When explaining water's properties, always link the molecular structure (polarity and hydrogen bonds) to the biological significance. A common mistake is describing properties without explaining the underlying mechanism. For example: "Water has a high specific heat capacity *because* many hydrogen bonds must be broken before the temperature rises, *which means* aquatic environments remain stable for enzyme activity."

Real-World Application: Hypothermia treatment in hospitals relies on water's high specific heat capacity — warming intravenous fluids before infusion prevents further heat loss. Conversely, athletes use evaporative cooling (sweating) to prevent overheating, exploiting water's high latent heat of vaporisation.

1.2 Carbohydrates

Carbohydrate: An organic molecule containing carbon, hydrogen and oxygen, often in the ratio $C_n(H_2O)_n$. They function as energy sources (glucose), energy stores (starch, glycogen) and structural components (cellulose, chitin).

Monosaccharides

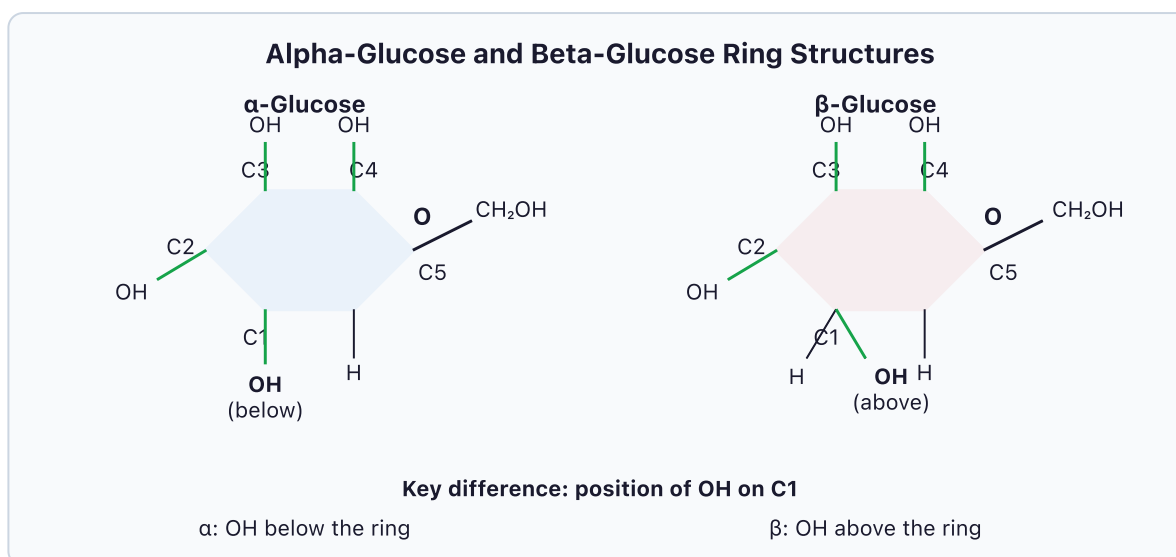
Monosaccharides are the simplest carbohydrates — single sugar units that cannot be hydrolysed further. They have the general formula $(CH_2O)_n$ and are classified by the number of carbon atoms:

- **Trioses (3C)** — e.g., glyceraldehyde, dihydroxyacetone phosphate (DHAP). These are key intermediates in glycolysis and the Calvin cycle
- **Pentoses (5C)** — e.g., **ribose** (found in RNA, ATP, NAD, FAD, coenzyme A) and **deoxyribose** (found in DNA, which lacks the $-OH$ on carbon-2)
- **Hexoses (6C)** — e.g., **glucose** (α and β isomers), fructose, galactose. Glucose is the primary respiratory substrate in most organisms

All monosaccharides are **reducing sugars** — they can donate electrons to other molecules. This is the basis of the Benedict's test.

Alpha-Glucose vs Beta-Glucose

Glucose exists in two isomeric ring forms. **Alpha-glucose (α)** has the hydroxyl group ($-\text{OH}$) on carbon-1 positioned **below** the plane of the ring. **Beta-glucose (β)** has it **above** the ring. This seemingly small difference has enormous consequences for the polysaccharides they form: α -glucose produces helical storage molecules (starch, glycogen), while β -glucose produces straight structural chains (cellulose).



Condensation and Hydrolysis

Monosaccharides join together by **condensation reactions**: the $-\text{OH}$ group of one monosaccharide reacts with the $-\text{OH}$ of another, releasing a molecule of water (H_2O) and forming a **glycosidic bond**. The reverse process, **hydrolysis**, uses water to break the glycosidic bond, catalysed by specific enzymes (e.g., maltase, lactase, sucrase). Digestion of carbohydrates is essentially hydrolysis of glycosidic bonds.

Disaccharides and Glycosidic Bonds

Disaccharide	Monosaccharides	Bond	Where Found
Maltose	Glucose + Glucose	α -1,4 glycosidic	Germinating seeds (starch digestion intermediate)
Sucrose	Glucose + Fructose	α -1,2 glycosidic	Transport sugar in phloem of plants; table sugar

Lactose	Glucose + Galactose	β -1,4 glycosidic	Mammalian milk; requires lactase for digestion
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Real-World Application: Lactose intolerance occurs when adults produce insufficient **lactase** enzyme to hydrolyse lactose. Undigested lactose is fermented by gut bacteria, producing gas and causing bloating, cramps, and diarrhoea. This is common in populations with historically low dairy consumption (e.g., East Asian, African). Lactase persistence (continued production in adults) evolved in populations with a long history of pastoral farming.

Polysaccharides

Polysaccharides are long chains of monosaccharides linked by glycosidic bonds. They are **not** reducing sugars (the anomeric carbons are involved in bonds). Their structure determines their function:

Polysaccharide	Monomer	Structure	Function	Key Features
Starch (amylose)	α -glucose	Unbranched helix; α -1,4 glycosidic bonds only	Long-term energy store in plants (tubers, seeds, starch grains in chloroplasts)	Compact helix; insoluble (no osmotic effect); iodine turns it blue-black by entering the helix
Starch (amylopectin)	α -glucose	Branched; α -1,4 bonds with α -1,6 branches every 24–30 residues	More rapidly mobilised energy in plants	Branches provide many terminal ends for enzyme action; faster hydrolysis than amylose

Glycogen	α -glucose	Highly branched; α -1,4 and α -1,6 bonds; more branches than amylopectin (every 8–12 residues)	Short-term energy store in animals (mainly liver and skeletal muscle)	Very many branch ends allow rapid glucose release; granules visible in liver cells; stored with associated water
Cellulose	β -glucose	Straight, unbranched chains; β -1,4 bonds; adjacent monomers rotated 180°	Structural component of plant cell walls	Chains form hydrogen bonds with neighbouring chains, creating strong microfibrils ; very high tensile strength; indigestible by most animals (no cellulase); provides dietary fibre

Exam Tip: Storage molecules (starch and glycogen) are ideal stores because they are: (1) **insoluble** — do not affect water potential or diffuse out of cells, (2) **compact** — many glucose units in small space, (3) easily hydrolysed to release glucose when needed. Cellulose has high **tensile strength** because of hydrogen bonds between parallel chains forming microfibrils — not because of the β -1,4 bonds alone.

Exam Tip: A common 6-mark question asks you to compare and contrast starch and cellulose. Remember: both are glucose polymers, but starch uses α -glucose with α -1,4 bonds (coiled) while cellulose uses β -glucose with β -1,4 bonds (straight). This structural difference accounts for their entirely different functions (storage vs structural).

1.3 Lipids

Lipid: A diverse group of hydrophobic (water-insoluble) organic molecules containing carbon, hydrogen and oxygen, with a higher H:O ratio than carbohydrates. Includes triglycerides, phospholipids, cholesterol and waxes. They are not true polymers — they do not form by condensation of repeating monomers in the same way proteins and polysaccharides do.

Triglycerides (Fats and Oils)

Triglycerides are formed from one molecule of **glycerol** (a 3-carbon alcohol) and three **fatty acid** chains, joined by **ester bonds** through three condensation reactions (releasing 3 H₂O). They are the main long-term energy storage molecules in animals and plants.

Glycerol + 3 Fatty acids → Triglyceride + 3H₂O (condensation / esterification)

Fatty acids consist of a hydrocarbon chain (typically 14–20 carbons long) with a carboxyl group (–COOH) at one end. They can be:

- **Saturated** — no C=C double bonds in the hydrocarbon chain; chains are straight and pack tightly together; solid at room temperature (e.g., animal fats: butter, lard). Associated with increased LDL cholesterol and cardiovascular disease risk
- **Monounsaturated** — one C=C double bond; introduces a kink in the chain; liquid at room temperature (e.g., olive oil)
- **Polyunsaturated** — two or more C=C double bonds; multiple kinks prevent tight packing; oils remain liquid even at low temperatures (e.g., sunflower oil, fish oils). Omega-3 fatty acids (in oily fish) are associated with reduced cardiovascular disease risk

Why Lipids Store More Energy Than Carbohydrates

Lipids release approximately **twice as much energy per gram** (37 kJ g⁻¹) compared to carbohydrates (16 kJ g⁻¹). This is because lipids have a much higher proportion of hydrogen atoms relative to oxygen atoms — the C–H bonds provide more energy when oxidised. Additionally, lipids are stored in a **anhydrous** (water-free) state, making them lighter per unit of energy. This makes them ideal for long-distance migrants (birds) and organisms that need to minimise weight.

Phospholipids

Phospholipids are structurally similar to triglycerides, but with one fatty acid replaced by a **phosphate group** (often with an additional polar group such as choline). This gives them an **amphipathic** (or amphiphilic) nature:

- **Hydrophilic head** — the phosphate group is polar and charged; interacts with water
- **Hydrophobic tails** — the two fatty acid chains are non-polar; repelled by water

In aqueous environments, phospholipids spontaneously arrange into a **bilayer** with hydrophilic heads facing outward (towards water) and hydrophobic tails facing inward (away from water). This is the fundamental structural basis of all biological membranes. One tail is usually saturated (straight) and one unsaturated (kinked), affecting membrane fluidity.

Cholesterol

Cholesterol is a lipid with a distinctive four-ring hydrocarbon structure (a steroid) with a short hydrocarbon tail and a hydroxyl group. In cell membranes, cholesterol molecules sit between phospholipid tails. Cholesterol plays a dual role in regulating membrane **fluidity**:

- **At high temperatures:** cholesterol reduces fluidity by restricting phospholipid movement and making the membrane less permeable to small water-soluble molecules
- **At low temperatures:** cholesterol prevents the membrane from becoming too rigid by disrupting regular packing of fatty acid tails, acting as a "fluidity buffer"

Cholesterol is also the precursor for steroid hormones (oestrogen, testosterone, cortisol), bile salts (emulsify fats in digestion) and vitamin D.

Real-World Application: Blood cholesterol is transported by **lipoproteins**. **LDL (low-density lipoprotein)** carries cholesterol to tissues and can deposit it in arterial walls (increasing atherosclerosis risk). **HDL (high-density lipoprotein)** transports cholesterol back to the liver for excretion. High LDL:HDL ratio is a risk factor for cardiovascular disease. Statins are drugs that lower LDL by inhibiting HMG-CoA reductase, the enzyme controlling cholesterol synthesis in the liver.

1.4 Proteins

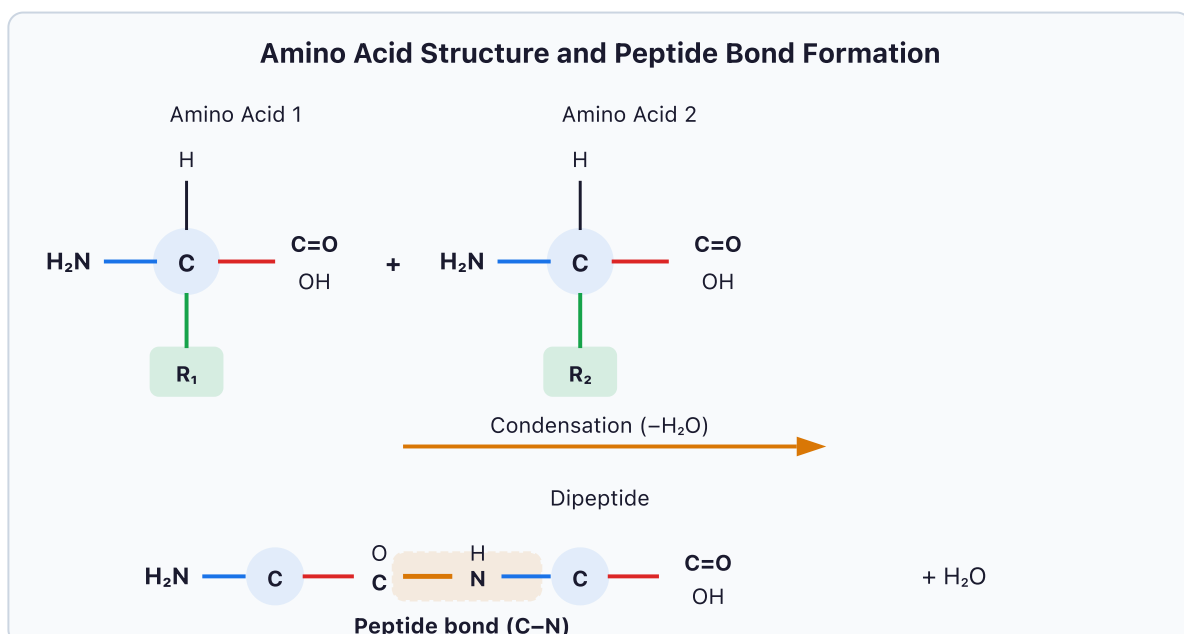
Protein: A polymer of amino acids joined by peptide bonds. Proteins have enormous structural and functional diversity: enzymes, antibodies, transport proteins (haemoglobin, channel proteins), structural proteins (collagen, keratin), hormones (insulin), receptors, motor proteins (actin, myosin) and storage proteins (casein in milk).

Amino Acid Structure

All 20 amino acids share a common core structure: a central **alpha carbon (α -C)** bonded to four groups:

- An **amino group** ($-\text{NH}_2$) — acts as a base (can accept H^+)
- A **carboxyl group** ($-\text{COOH}$) — acts as an acid (can donate H^+)
- A **hydrogen atom**
- A variable **R group** (side chain) — determines the properties of each amino acid (polar, non-polar, charged, aromatic, sulfur-containing)

At physiological pH (~ 7.4), amino acids exist as **zwitterions** with the amino group protonated ($-\text{NH}_3^+$) and the carboxyl group deprotonated ($-\text{COO}^-$).



Peptide Bonds and Polypeptides

Amino acids are linked by **peptide bonds** formed through condensation reactions between the amino group ($-\text{NH}_2$) of one amino acid and the carboxyl group ($-\text{COOH}$) of another, releasing water. The bond formed is C–N (with partial double bond character due to resonance). A chain of two amino acids is a dipeptide; many amino acids form a **polypeptide**. Peptide bonds are broken by **hydrolysis** (adding water), catalysed by protease enzymes during digestion.

Levels of Protein Structure

Level	Description	Bonds/Forces Involved	Example
Primary	The specific linear sequence of amino acids in the polypeptide chain, determined by the gene. Even one amino acid change can alter function (e.g., sickle cell anaemia: Glu → Val at position 6)	Peptide bonds (covalent)	Insulin: chain A (21 aa) and chain B (30 aa)
Secondary	Regular local folding into α-helices (right-handed coil) or β-pleated sheets (zigzag pattern). These arise from hydrogen bonding along the polypeptide backbone	Hydrogen bonds between C=O and N–H groups of the peptide backbone (every 4th amino acid in α -helix)	Keratin is largely α -helical; silk fibroin has extensive β -sheets
Tertiary	The overall 3D shape of a single polypeptide chain, determined by interactions between R groups. This gives the protein its specific shape and function	Hydrogen bonds (between polar R groups), ionic bonds (between charged R groups), disulfide bridges (covalent S–S bonds between cysteine residues), hydrophobic interactions (non-polar R groups cluster in the interior,	Myoglobin, enzymes like lysozyme

	(e.g., the active site of an enzyme)	away from water), and van der Waals forces	
Quaternary	The arrangement of two or more polypeptide subunits and/or prosthetic groups (non-protein components) into a functional protein complex	Same as tertiary (between subunits): hydrogen bonds, ionic bonds, disulfide bridges, hydrophobic interactions	Haemoglobin (4 subunits + 4 haem groups); collagen (3 polypeptide chains wound together); antibodies (2 heavy + 2 light chains)

Globular vs Fibrous Proteins

Feature	Globular Proteins	Fibrous Proteins
Shape	Spherical/compact; polypeptide chain folded into a ball	Long, elongated; chains arranged in parallel or coiled
Solubility	Generally soluble in water (hydrophilic R groups on surface)	Generally insoluble in water
Function	Metabolic/functional roles: enzymes, antibodies, haemoglobin, hormones, receptors	Structural roles: support, protection, tensile strength
Examples	Haemoglobin, insulin, immunoglobulins, lysozyme, catalase	Collagen (connective tissue, tendons, bone matrix), keratin (hair, nails, skin), elastin (lungs, arteries)

Exam Tip: For a 6-mark question on protein structure, make sure you can describe ALL four levels and name the specific bonds at each level. Remember: primary structure (gene-determined sequence) determines all higher levels — changing the primary structure can alter folding and function. Denaturation means loss of tertiary/quaternary structure but NOT breaking of peptide bonds (primary structure intact).

Real-World Application: Sickle cell anaemia is caused by a single amino acid substitution in the β -globin chain of haemoglobin (glutamic acid $\rightarrow$ valine at position 6). This changes a hydrophilic amino acid to a hydrophobic one, causing haemoglobin molecules to aggregate in low oxygen conditions, distorting red blood cells into a sickle shape. This demonstrates how primary structure determines function. Heterozygous carriers have sickle cell trait and partial resistance to malaria — an example of heterozygote advantage maintaining a harmful allele in populations.

1.5 Enzymes

Enzyme: A biological catalyst, usually a globular protein, that lowers the **activation energy** (E_a) of a specific metabolic reaction, greatly increasing the rate of reaction without being consumed or permanently altered. Enzymes typically increase reaction rates by a factor of 10^6 to 10^{12} .

Enzymes work by providing an alternative reaction pathway with a lower activation energy. They do NOT change the equilibrium position or the overall free energy change (ΔG) of the reaction — they just help the reaction reach equilibrium faster.

Lock and Key Model vs Induced Fit Model

The **lock and key model** (Emil Fischer, 1894) proposes that the substrate has a shape exactly complementary to the enzyme's **active site**, like a key fitting into a lock. While this explains enzyme specificity, it is an oversimplification.

The **induced fit model** (Daniel Koshland, 1958) is more accurate. It proposes that the active site is not a rigid structure but changes shape slightly when the substrate binds, moulding around the substrate to form a tighter, more precise fit. This conformational change:

- Places strain on bonds within the substrate, weakening them and lowering activation energy
- Brings catalytic amino acid residues in the active site into optimal positions for catalysis
- May exclude water from the active site, creating a more favourable chemical environment

Factors Affecting Enzyme Activity

- **Temperature:** Increasing temperature increases kinetic energy of molecules, leading to more frequent and energetic enzyme-substrate collisions. Rate roughly doubles for every 10°C rise ($Q_{10} \approx 2$) up to the **optimum temperature**. Above the optimum, hydrogen bonds and ionic bonds maintaining the tertiary structure break, the active site changes shape (**denaturation**), and the rate drops sharply to zero. Denaturation is usually permanent (irreversible). Human enzymes typically have optimum ~37°C; thermophilic bacteria (e.g., *Thermus aquaticus*) have enzymes with optima of 70–80°C
- **pH:** Each enzyme has a characteristic optimum pH (e.g., pepsin pH 2, trypsin pH 8, most intracellular enzymes pH 7). Changes in pH alter the ionisation of amino acid R groups, disrupting ionic bonds and hydrogen bonds in the tertiary structure. Extreme pH causes denaturation. The active site shape changes and can no longer bind the substrate
- **Substrate concentration:** At low [S], rate increases proportionally (first-order kinetics) as more substrate molecules are available to bind active sites. Eventually, all active sites are occupied (enzyme saturated), and rate reaches **$V_{\max}$** (zero-order kinetics). Further substrate increases have no effect unless more enzyme is added
- **Enzyme concentration:** If substrate is in excess, increasing enzyme concentration increases the rate proportionally — more active sites are available for catalysis

Enzyme Inhibition

Type	Mechanism	Effect on $V_{\max}$	Effect on K_m	Reversibility	Example
Competitive	Inhibitor has a similar shape to substrate; competes for the active site; blocks substrate binding	Unchanged (can be overcome with excess substrate)	Increased (apparent lower affinity)	Reversible	Malonate inhibits succinate dehydrogenase (competes with succinate); statins compete with HMG-CoA
Non-competitive	Inhibitor binds to an	Reduced (fewer	Unchanged (substrate	Can be reversible or	Heavy metal ions (Hg^{2+} , Pb^{2+}) bind to

	allosteric site (not the active site); causes conformational change that distorts the active site shape; substrate can still bind but no catalysis occurs	functional enzyme-substrate complexes)	still binds normally)	irreversible	sulfhydryl groups; cyanide inhibits cytochrome c oxidase; nerve agents inhibit acetylcholinesterase
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Michaelis-Menten equation: $v = V_{\max} \times [S] / (K_m + [S])$

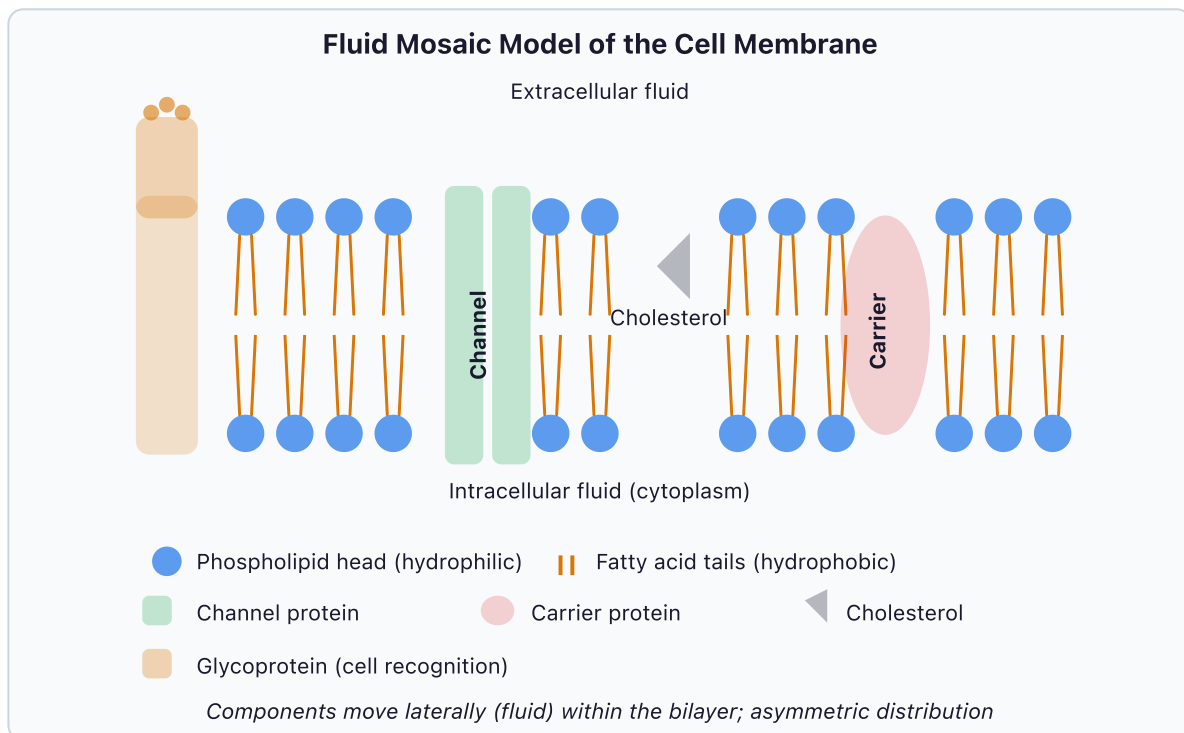
Where **K_m** (Michaelis constant) is the substrate concentration at which the reaction rate is half $V_{\max}$. A low K_m indicates high enzyme-substrate affinity (the enzyme reaches half its maximum rate at a low substrate concentration).

Exam Tip: On Lineweaver-Burk plots ($1/v$ vs $1/[S]$): competitive inhibition changes the x-intercept (K_m appears to increase) but the y-intercept ($1/V_{\max}$) stays the same. Non-competitive inhibition changes the y-intercept ($V_{\max}$ decreases) but the x-intercept (K_m) stays the same. Know how to interpret both types of graph.

Real-World Application: Drug design exploits enzyme inhibition. Penicillin is a competitive inhibitor of transpeptidase, preventing bacterial cell wall synthesis. HIV protease inhibitors block the viral protease needed to produce functional viral proteins. ACE inhibitors treat hypertension by inhibiting angiotensin-converting enzyme. Understanding K_m and $V_{\max}$ helps pharmacologists design effective drug dosages.

1.6 Cell Membranes

Fluid mosaic model (Singer and Nicolson, 1972): The accepted model of membrane structure describing a **phospholipid bilayer** with various proteins, cholesterol, and glycolipids/glycoproteins distributed throughout like tiles in a mosaic. The "fluid" aspect refers to the fact that phospholipids and some proteins can move laterally within the bilayer. The "mosaic" refers to the scattered arrangement of different protein molecules.



Membrane Components and Functions

Component	Function	Additional Detail
Phospholipid bilayer	Barrier to water-soluble (hydrophilic) substances; provides the basic structure and fluidity	The hydrophobic core prevents ions, polar molecules and large molecules from crossing freely; small non-polar molecules (O_2 , CO_2 , steroid hormones) can diffuse through
Channel proteins	Allow specific ions/small polar molecules to pass through a water-filled pore (passive, down concentration gradient)	Highly selective for particular ions (Na^+ , K^+ , Ca^{2+}); some are gated (open/close in response to voltage, ligands or mechanical stimuli)

Carrier proteins	Transport specific molecules across the membrane by changing shape	Used in both facilitated diffusion (passive) and active transport (requires ATP). Example: Na ⁺ /K ⁺ ATPase, GLUT transporters for glucose
Glycoproteins	Cell signalling, cell recognition, immune response (antigens)	Sugar chains on the extracellular surface act as receptors for hormones and neurotransmitters; form the glycocalyx; determine blood groups (ABO antigens); enable cell-cell adhesion
Glycolipids	Cell recognition, cell adhesion, membrane stability	Important in tissue formation; act as antigens on cell surfaces; contribute to the glycocalyx
Cholesterol	Regulates membrane fluidity and permeability	Reduces membrane permeability to small hydrophilic molecules; prevents crystallisation of fatty acid tails at low temperatures; maintains membrane stability across a temperature range

Factors Affecting Membrane Permeability

Temperature increases cause phospholipids to have more kinetic energy and move more, increasing membrane fluidity and permeability. At very high temperatures (above ~45°C), membrane proteins denature and phospholipids move so much that temporary gaps form, dramatically increasing permeability. Organic solvents (e.g., ethanol, acetone) dissolve lipids in the membrane, disrupting the bilayer structure and increasing permeability.

This can be investigated using **beetroot discs**: placing them in water at different temperatures and measuring the leakage of the red pigment **betacyanin** using a colorimeter. Higher temperature = more pigment leakage = higher absorbance reading.

Exam Tip: When explaining the beetroot experiment, describe the mechanism precisely: high temperatures denature the **intrinsic proteins** in the tonoplast (vacuole membrane) and cell membrane, and increase phospholipid movement, creating gaps. Betacyanin leaks from the vacuole through the damaged tonoplast, then out through the damaged cell membrane into

the surrounding water. This is NOT osmosis — it is increased permeability due to membrane disruption.

1.7 Transport Across Membranes

Mechanism	Energy Required?	Description	Examples
Simple diffusion	No (passive)	Net movement of molecules/ions from a region of higher concentration to lower concentration, down a concentration gradient. Molecules must be small and non-polar (or very small and uncharged) to cross the phospholipid bilayer directly	O ₂ and CO ₂ across alveolar epithelium; steroid hormones crossing cell membranes; ethanol absorption in the gut
Facilitated diffusion	No (passive)	Movement of ions or large/polar molecules down a concentration gradient through specific channel proteins or carrier proteins . The proteins are highly specific (each transports only one type or a closely related group of molecules)	Glucose via GLUT transporters; Na ⁺ , K ⁺ , Cl ⁻ through ion channels; aquaporins for water
Osmosis	No (passive)	Net movement of water molecules from a region of higher water potential (Ψ) to a region of lower water potential across a partially permeable membrane . Water moves to dilute the more concentrated solution	Water absorption in root hair cells; water reabsorption in kidney tubules; red blood cells in hypotonic/hypertonic solutions

Active transport	Yes (ATP)	Movement of molecules/ions against the concentration gradient (from low to high concentration) using carrier proteins that are powered by ATP hydrolysis. The carrier changes shape to move the substance across	Na ⁺ /K ⁺ ATPase (3 Na ⁺ out, 2 K ⁺ in); mineral ion uptake by root cells; glucose reabsorption in kidney PCT (co-transport with Na ⁺)
Endocytosis	Yes (ATP)	Cell engulfs extracellular material by infolding of the membrane to form a vesicle. Phagocytosis = engulfing solid particles; pinocytosis = engulfing liquid/dissolved substances	White blood cells engulfing bacteria (phagocytosis); uptake of LDL cholesterol by receptor-mediated endocytosis
Exocytosis	Yes (ATP)	Vesicles (from Golgi or other sources) fuse with the cell surface membrane, releasing their contents outside the cell	Secretion of hormones (insulin from beta cells); neurotransmitter release at synapses; secretion of digestive enzymes

Water potential (Ψ): The tendency of water molecules to move from one region to another by osmosis. Pure water has the highest water potential, defined as **0 kPa**. Adding solute lowers (makes more negative) the water potential. Water always moves from **less negative** (higher) Ψ to **more negative** (lower) Ψ .

$$\Psi = \Psi_s + \Psi_p$$

Where Ψ_s = solute potential (always negative; more solute = more negative) and Ψ_p = pressure potential (positive in turgid plant cells due to wall pressure; zero in animal cells).

Osmosis and Cells

Solution Type	Water Potential Relative to Cell	Animal Cell Response	Plant Cell Response
Hypotonic (dilute; higher Ψ than cell)	$\Psi_{\text{solution}} > \Psi_{\text{cell}}$	Water enters by osmosis; cell swells and may burst (lysis / haemolysis in red blood cells)	Water enters; cell becomes turgid (firm); cell wall prevents bursting; turgor pressure builds up
Isotonic (same Ψ)	$\Psi_{\text{solution}} = \Psi_{\text{cell}}$	No net water movement; cell stays normal	No net water movement; flaccid (soft, not turgid)
Hypertonic (concentrated; lower Ψ than cell)	$\Psi_{\text{solution}} < \Psi_{\text{cell}}$	Water leaves by osmosis; cell shrinks (crenation)	Water leaves; cytoplasm shrinks away from cell wall (plasmolysis); membrane peels away; cell becomes flaccid then plasmolysed

Exam Tip: Osmosis questions are very common. Always state: water moves from **higher** (less negative) water potential to **lower** (more negative) water potential. Never say "water moves from dilute to concentrated" — use correct terminology. In calculations, remember that Ψ_s is always negative and Ψ_p can be positive (plants) or zero (animal cells). At incipient plasmolysis, $\Psi_p = 0$ so $\Psi_{\text{cell}} = \Psi_s$.

1.8 The Cardiovascular System

Blood Vessel Structure and Function

All blood vessels (except capillaries) have walls with three layers: **tunica intima** (endothelium lining), **tunica media** (smooth muscle and elastic tissue), and **tunica adventitia** (outer connective tissue). The relative thickness and composition of these layers varies:

Feature	Artery	Arteriole	Capillary	Venule	Vein
Wall thickness	Thick	Medium	One cell thick (endothelium only)	Thin	Thin (relative to lumen)
Lumen diameter	Relatively small	Very small	Very narrow (just wide enough for one RBC to pass)	Small	Large
Elastic tissue	Abundant (stretches and recoils with each heartbeat, maintaining smooth blood flow)	Little	None	Little	Some
Smooth muscle	Thick layer	Relatively thick (controls diameter for vasoconstriction/vasodilation)	None	Very little	Thin layer
Valves	No (except semilunar valves at base of aorta and pulmonary artery)	No	No	Sometimes	Yes (prevent backflow; especially in limbs where blood flows against gravity)

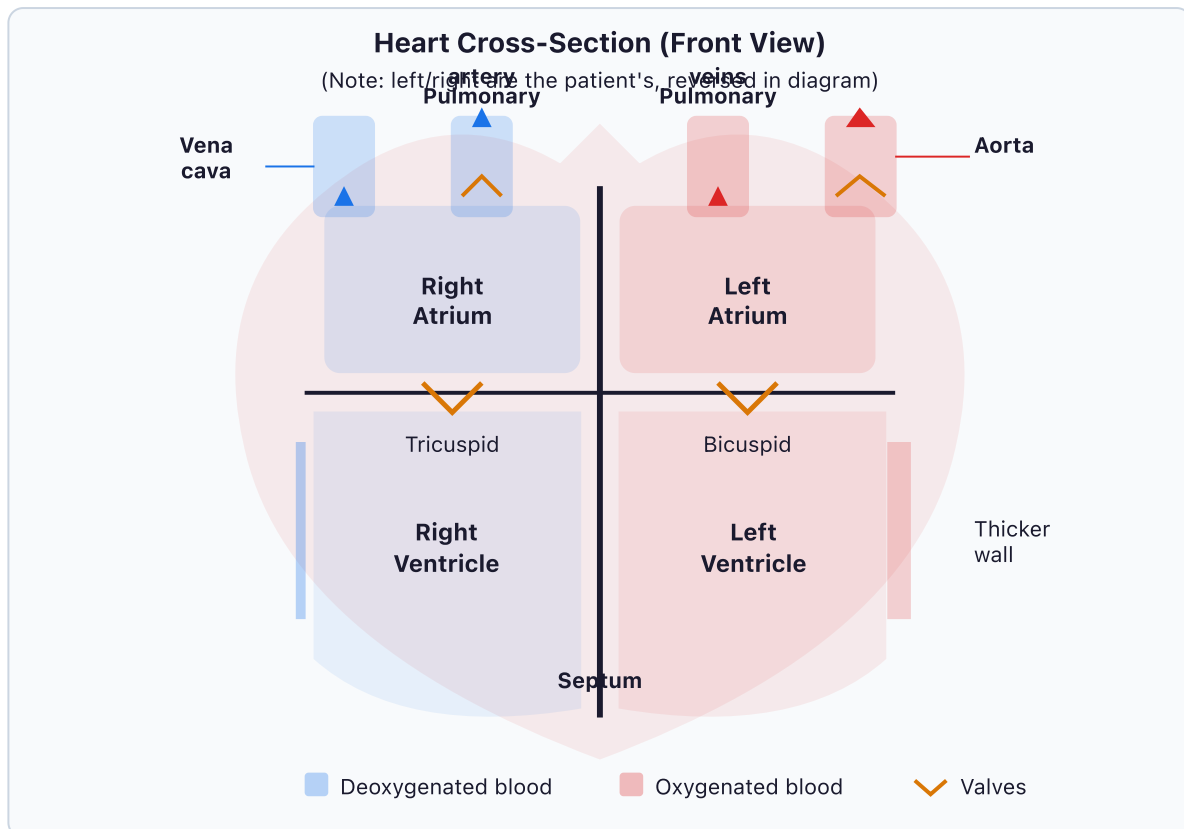
Blood pressure	High, pulsatile	Decreasing	Low, steady	Very low	Very low
Function	Carry blood away from heart under high pressure	Control blood flow to capillary beds; regulate blood pressure	Exchange of substances between blood and tissues (thin walls + large SA)	Collect blood from capillaries	Return blood to heart under low pressure

The Heart Structure

The heart is a muscular double pump with four chambers. The **right side** pumps deoxygenated blood to the lungs via the pulmonary artery (pulmonary circulation). The **left side** pumps oxygenated blood to the body via the aorta (systemic circulation). The left ventricle wall is approximately **three times thicker** than the right because it must generate much higher pressure to push blood around the entire body, while the right ventricle only needs to push blood to the nearby lungs.

Key structural features:

- **Septum** — separates left and right sides, preventing mixing of oxygenated and deoxygenated blood
- **Atrioventricular (AV) valves:** tricuspid (right) and bicuspid/mitral (left) — prevent backflow from ventricles into atria during ventricular systole
- **Semilunar valves** — in the aorta and pulmonary artery; prevent backflow into ventricles during diastole
- **Tendinous cords (chordae tendineae)** — prevent AV valves from inverting under pressure
- **Coronary arteries** — branch from the aorta to supply heart muscle (cardiac muscle) with oxygen and glucose for continuous aerobic respiration



The Cardiac Cycle

The cardiac cycle is the sequence of events during one complete heartbeat, lasting approximately 0.8 seconds at rest (75 bpm):

1. **Atrial systole (0.1 s):** Both atria contract simultaneously, increasing atrial pressure. Blood is pushed through the open AV valves into the ventricles (topping them up — ventricles are already ~70% full from passive filling during diastole). Semilunar valves remain closed
2. **Ventricular systole (0.3 s):** Both ventricles contract from the apex upward. Ventricular pressure rises sharply, causing AV valves to snap shut (producing the first heart sound, "lub"). When ventricular pressure exceeds arterial pressure, semilunar valves open and blood is ejected into the aorta and pulmonary artery
3. **Diastole (0.4 s):** All chambers relax. Ventricular pressure drops below arterial pressure; semilunar valves close (producing the second heart sound, "dub"), preventing backflow. Pressure in the atria exceeds ventricular pressure, so AV valves open and blood flows passively from the veins into the atria and then into the ventricles

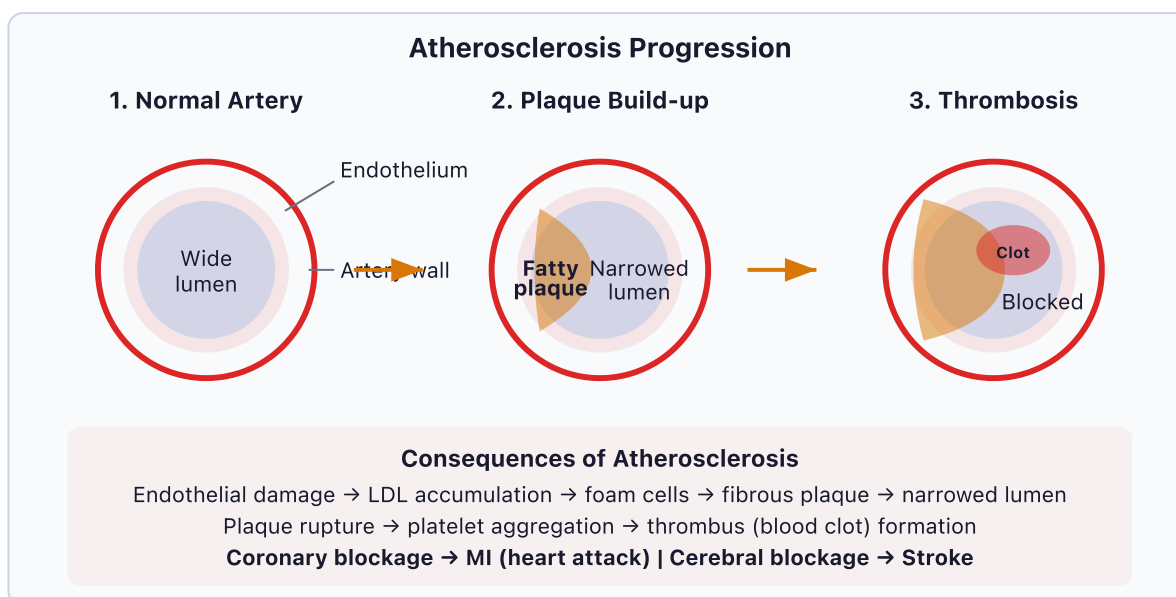
Exam Tip: In cardiac cycle pressure-time graphs, key things to identify: (1) atrial pressure rises slightly during atrial systole, (2) ventricular pressure rises steeply during ventricular

systole, (3) the AV valves open when atrial pressure exceeds ventricular pressure and close when ventricular pressure exceeds atrial pressure, (4) semilunar valves open when ventricular pressure exceeds aortic/pulmonary pressure. The volume of blood in the ventricle decreases during systole and increases during diastole.

Atherosclerosis and Cardiovascular Disease (CVD)

Atherosclerosis is the progressive build-up of fatty deposits (**atheromas/plaques**) within the walls of arteries. The process occurs over decades:

1. **Endothelial damage** — caused by high blood pressure, smoking toxins, high LDL cholesterol, or infection. Damage to the endothelial lining triggers an inflammatory response
2. **LDL accumulation** — low-density lipoproteins infiltrate the damaged arterial wall and become oxidised. Macrophages engulf the oxidised LDL, becoming **foam cells**
3. **Plaque formation** — foam cells, smooth muscle cells, cholesterol crystals and calcium deposits accumulate beneath the endothelium, forming a fibrous plaque with a lipid-rich core
4. **Narrowing of lumen** — the plaque protrudes into the arterial lumen, reducing blood flow and increasing blood pressure
5. **Plaque rupture and thrombosis** — if the fibrous cap ruptures, the exposed collagen activates the clotting cascade. Platelets aggregate at the site, forming a **thrombus (blood clot)**
6. **Consequences** — if a coronary artery is blocked: **myocardial infarction** (heart attack) due to oxygen deprivation of cardiac muscle. If a cerebral artery is blocked: **stroke**. An embolus is a detached clot that can travel and block a vessel elsewhere



Risk Factors for CVD

Risk Factor	How It Increases CVD Risk	Modifiable?
Smoking	Carbon monoxide reduces oxygen-carrying capacity; nicotine increases heart rate and blood pressure; chemicals damage endothelium; increases platelet stickiness	Yes
High blood cholesterol (high LDL)	More LDL available to infiltrate and oxidise in arterial walls; accelerates plaque formation	Yes (diet, statins)
High blood pressure (hypertension)	Damages endothelial lining; increases risk of plaque rupture; puts strain on heart	Yes (exercise, diet, medication)
Obesity	Associated with high blood pressure, high LDL, type 2 diabetes, and chronic inflammation	Yes
Physical inactivity	Reduces HDL; increases blood pressure and obesity risk; reduces cardiovascular fitness	Yes
High-salt diet	Increases blood volume (water retention) and blood pressure	Yes

Diabetes	High blood glucose damages blood vessel walls; often associated with other risk factors	Partially (type 2 manageable with lifestyle)
Genetic predisposition	Family history of CVD; inherited conditions such as familial hypercholesterolaemia	No
Age and sex	Risk increases with age; males at higher risk until post-menopause (oestrogen is protective)	No

Exam Tip: In questions about CVD risk factors, explain the **mechanism** by which each factor increases risk, not just that it does. For example, do not just say "smoking causes heart disease" — explain that chemicals in smoke damage the endothelium, nicotine raises heart rate and blood pressure, and carbon monoxide reduces oxygen transport. Also note that risk factors are often **correlated** rather than necessarily causative (this distinction is important in evaluating epidemiological studies).

1.9 Gas Exchange in the Lungs

Gas exchange surface: A specialised area where respiratory gases (O₂ and CO₂) are exchanged between an organism and its environment by diffusion down concentration gradients. All gas exchange surfaces share common adaptations to maximise the rate of diffusion.

Fick's Law of Diffusion

Rate of diffusion $\propto$ (surface area $\times$ concentration difference) / thickness of exchange surface

This relationship explains why gas exchange surfaces are adapted to have a **large surface area**, maintain a **steep concentration gradient**, and have a **short diffusion distance**.

Adaptations of the Mammalian Lung for Efficient Gas Exchange

- **Large surface area (~70 m²)** — provided by approximately 300–500 million alveoli (tiny air sacs, each 200–300 µm diameter), which collectively provide a surface area roughly the size of a tennis court
- **Thin barrier (short diffusion distance)** — alveolar walls are one cell thick (squamous epithelium, ~0.2 µm); capillary walls are also one cell thick. Total diffusion distance from air to blood is only about 0.5 µm
- **Rich blood supply** — each alveolus is surrounded by a dense capillary network. Blood constantly carries away O₂ (maintaining low O₂ concentration in blood) and delivers CO₂ (maintaining high CO₂ in blood). This maintains steep concentration gradients for both gases
- **Ventilation** — continuous breathing replaces air in the alveoli, maintaining high O₂ and low CO₂ in the alveolar air space, keeping the concentration gradient steep
- **Moist surface** — alveolar lining is covered by a thin film of moisture; gases dissolve before diffusing across. Surfactant (a phospholipid mixture secreted by type II pneumocytes) reduces surface tension, preventing alveoli from collapsing
- **Elastin fibres** in alveolar walls allow elastic recoil during expiration

Lung Structure and Airway Hierarchy

Air passes through: **trachea** → **bronchi** (left and right) → **bronchioles** (smooth muscle in walls; can constrict in asthma) → **terminal bronchioles** → **respiratory bronchioles** → **alveolar ducts** → **alveoli**.

The trachea and bronchi are supported by **C-shaped cartilage rings** that prevent collapse during inhalation (when internal pressure drops) while allowing the oesophagus behind to expand during swallowing. **Goblet cells** in the airways secrete mucus that traps inhaled pathogens, dust, and particles. **Ciliated epithelial cells** beat in coordinated waves (mucociliary escalator) to move the mucus and trapped debris upward towards the throat, where it is swallowed.

Real-World Application: Smoking and lung disease: Tar in cigarette smoke destroys cilia and stimulates mucus overproduction, leading to chronic bronchitis (persistent cough). Toxins cause inflammation and destruction of alveolar walls, reducing surface area — this is **emphysema** (part of COPD). Together, loss of surface area, thickening of diffusion barrier,

and impaired ventilation severely reduce gas exchange efficiency. Carcinogens in tar can cause mutations in genes controlling cell division, leading to **lung cancer**.

Pulmonary Ventilation

Pulmonary ventilation rate ($\text{dm}^3 \text{ min}^{-1}$) = tidal volume (dm^3) × breathing rate (breaths min^{-1})

Inspiration (inhalation):

- External intercostal muscles **contract**, pulling the rib cage up and out
- Diaphragm **contracts** and flattens (moves downward)
- Thoracic volume **increases**, intrapulmonary pressure **decreases** below atmospheric pressure
- Air is drawn into the lungs down the pressure gradient (Boyle's law)
- This is an **active** process (requires ATP for muscle contraction)

Expiration (exhalation) at rest:

- External intercostal muscles and diaphragm **relax**
- Elastic recoil of lungs and chest wall decreases thoracic volume
- Intrapulmonary pressure **increases** above atmospheric pressure
- Air is forced out of the lungs
- This is largely a **passive** process at rest (elastic recoil, no muscle contraction needed)

Forced expiration (during exercise) involves active contraction of **internal intercostal muscles** (pull ribs down and in) and **abdominal muscles** (push diaphragm up), increasing the rate and volume of air expelled.

Exam Tip: Always refer to Fick's law when answering questions about gas exchange efficiency. For every adaptation you describe, explicitly link it to one of the three factors: surface area, concentration gradient, or diffusion distance. Common mistake: saying "thick walls" instead of "thin walls" or confusing the role of cartilage rings (support, NOT gas exchange).

1.10 Biochemical Tests for Biological Molecules

Test	Reagent(s)	Procedure	Positive Result	Tests for
Benedict's test	Benedict's reagent (copper(II) sulfate in alkaline solution)	Add Benedict's reagent to sample; heat in water bath at 80°C for 2–5 minutes	Colour change: blue → green → yellow → orange → brick-red precipitate (Cu ₂ O). Semi-quantitative: more sugar = more red	Reducing sugars (all monosaccharides + some disaccharides: maltose, lactose)
Non-reducing sugar test	HCl, NaHCO ₃ , Benedict's reagent	(1) First test with Benedict's — must be negative . (2) Boil sample with dilute HCl (hydrolyses glycosidic bond). (3) Neutralise with NaHCO ₃ . (4) Re-test with Benedict's	Colour change to brick-red ONLY after hydrolysis step	Non-reducing sugars (sucrose — the anomeric carbons are locked in the glycosidic bond)
Iodine test	Iodine/potassium iodide (I ₂ /KI) solution	Add drops of iodine solution to sample at room temperature	Blue-black colour (iodine molecules become trapped inside the amylose helix, changing the light absorption)	Starch

Biuret test	NaOH (sodium hydroxide) then CuSO_4 (copper sulfate)	Add NaOH to make solution alkaline, then add dilute CuSO_4 drop by drop. Do NOT heat	Purple/lilac colour (Cu^{2+} ions form a complex with peptide bonds — at least two peptide bonds needed, so individual amino acids give negative result)	Proteins (specifically peptide bonds)
Emulsion test	Ethanol, then distilled water	Dissolve sample in ethanol (shake vigorously); pour into water in a clean test tube	White, cloudy emulsion (milky layer). Lipids are insoluble in water and form tiny suspended droplets that scatter light	Lipids (fats and oils)

Exam Tip: For the non-reducing sugar test, you MUST: (1) first perform Benedict's test and get a **negative result** (if positive, reducing sugars are present), (2) THEN hydrolyse with HCl, (3) neutralise with NaHCO_3 (essential — Benedict's only works in alkaline conditions), and (4) re-test with Benedict's. If positive only after hydrolysis, the sample contains non-reducing sugars (sucrose). Common mistake: forgetting to state the neutralisation step or not testing with Benedict's first.

Exam Tip: For quantitative analysis, use a **colorimeter** to measure absorbance/transmission of the colour change, then compare to a calibration curve made with known concentrations. This is more objective and accurate than judging colour by eye. Serial dilutions can be used to create the standard curve.

Real-World Application: These biochemical tests are used routinely in **clinical diagnostics** and **food science**. Urine dipstick tests for glucose (using glucose oxidase, a modified version of the reducing sugar concept) help diagnose diabetes mellitus. The biuret test principle is used in clinical protein assays (e.g. the BCA assay). Food labelling regulations require accurate determination of sugar, protein, and fat content in processed foods.